Alert Storm Protection

An optional per-workflow cap on daily notifications, with a controlled exception for severe deterioration. Disabled by default.

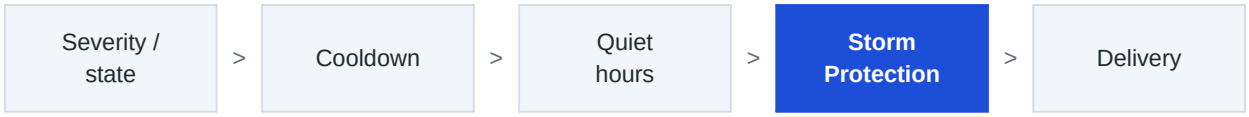
Problem

A persistent problem can stay Triggered or Critical for hours. Cooldown prevents constant alerts, but once it expires the same unchanged problem sends another reminder, all day. Each alert is correct; together they are noise.

Solution

Storm Protection adds a **daily quota** on ordinary notifications per tenant-local calendar day. When the quota is exhausted, an **exceptional alert** can still get through if the metric becomes severely bad, once per day per mechanism. Reminders are kept, not removed.

Alert flow



Storm Protection is the last gate. It only decides whether a notification that cooldown and quiet hours already allow is delivered. It does not replace or change state classification, cooldown, quiet hours, escalation or delivery.

If the quota blocks a notification, the workflow's state still changes normally and the finding is still recorded. Only the notification is held, and no cooldown starts.

Daily Quotas

Per workflow, per tenant, per tenant-local calendar day.

Limit	Applies to notifications whose resulting state is
Triggered	TRIGGERED (initial triggers, reminders, de-escalations)
Critical	CRITICAL (escalations, reminders)
Total	Either state

Each limit is a whole number of at least 1, or unlimited. A notification needs room under its state's limit and the Total limit.

Tenant-local reset

- Counters reset at the tenant's local midnight (Asia/Kolkata: midnight India time; America/New_York: midnight New York time).
- Not UTC midnight. Not a rolling 24-hour window.

What counts

Consumes quota	Does not consume quota
Delivered notifications, including partial delivery (at least one channel got through)	Held by cooldown or quiet hours Blocked by the daily limit No notification candidate Failed send where nobody received it Run with no notifier on its path All recipients muted or paused

Example

```
Critical daily limit = 3
Alert 1 -> sent Alert 2 -> sent Alert 3 -> sent
Alert 4 -> blocked by daily limit (workflow stays CRITICAL)
```

Exceptional Alerts

An exceptional alert bypasses an exhausted daily quota because an exceptional condition was met.

	Lower-is-worse metric	Higher-is-worse metric
Absolute exception current value reached the extreme threshold	current <= threshold	current >= threshold
Difference exception deteriorated enough from the previous valid finding	previous - current >= difference	current - previous >= difference

Direction comes from the workflow's existing thresholds. With no previous valid finding, the difference exception cannot trigger; the absolute exception still can.

Once-per-day latches

- Each mechanism has its own latch, so each can bypass the quota at most once per day.
- A latch is consumed **only when the exception is used to bypass an exhausted quota**. A true condition while quota remains sends an ordinary alert and consumes nothing.
- Latches reset at tenant-local midnight, with the counters.
- Latches and counters are runtime state, not configuration.

Combined exceptions

If both conditions qualify in the same run, **one** notification is sent, both reasons are recorded, and both latches are consumed.

Counters

An exceptional notification still increments the Total count and its resulting-state count, so usage can exceed the configured limit (for example 4 of 3). It keeps its normal reason, is treated as Critical severity, and the resulting state is unchanged.

Rules and Examples

How Storm Protection interacts with the rest of the engine.

Topic	Behavior
Cooldown	Runs first. Not bypassed by an exceptional condition. Escalations still ignore cooldown as before.
Quiet hours	Runs first. Holds every notification, Critical included, and is not bypassed by an exceptional condition. A direct Normal to Critical escalation is held too; no cooldown starts.
Escalation	Still subject to the daily quota unless it qualifies for an exceptional bypass.
Delivery failure	If nobody received it, quota and latch are handed back. Partial delivery counts as delivered.
Replay / concurrency	Replays do not count twice. Concurrent executions cannot consume the same final quota slot or latch twice.
Backward compatibility	Disabled by default. Workflows without the configuration behave exactly as before.

Example day

```
Limits 3 / 3 / 6; absolute -50; difference 20
09:00-14:00 six notifications sent (3 triggered + 3 critical)
15:00 -34 quota exhausted, no exception -> suppressed
16:00 -51 absolute exception, latch unused -> exceptional alert, latch used
17:00 -60 absolute latch already used -> suppressed
18:00 -80 deterioration 20 (>= 20), latch unused -> exceptional alert, latch used
00:00 (tenant local) counters and latches reset
```

Configuration

Per workflow, in the workflow builder under Alert state > Storm protection: enabled; daily limits (Triggered, Critical, Total, each a number or unlimited); exceptional alerts enabled; absolute value (on/off, number); difference (on/off, number greater than 0).